

Level 3: Advanced – Submission Report

Project: TaskFlow – Full Stack Task Management System

Organization:	Cognifyz Technologies	Domain:	Full Stack Web Development
Status:	Completed & 100% Verified	Author / Intern:	Full Stack Intern (vikast2849)
Tasks Included:	Task 5 & Task 6	Repository:	github.com/vikast2849/taskflow

1. Level Overview & Objectives

- **Task 5 (API Integration & Front-End Interaction):** Complete RESTful API (GET, POST 201, PUT 200, DELETE 200) with uniform JSON envelopes, and asynchronous fetch() UI client with loading spinners and error alerts.
- **Task 6 (Database Integration & User Authentication):** Persistent MongoDB integration via Mongoose, User & Task models, bcrypt 10-round password hashing, session/JWT auth, and strict multi-user authorization checks (403 Forbidden).

2. Route Endpoints & Verification URLs

- **Task 5 RESTful API UI:** /tasks/task-5 – Asynchronous frontend client communicating with Express REST API.
- **Task 5 REST API Endpoints:** /api/tasks – REST CRUD endpoints returning structured JSON envelopes.
- **Task 6 Auth Workspace:** /tasks/task-6 – Authenticated database workspace displaying persistent MongoDB user tasks.
- **Production Dashboard:** /dashboard – Unified task management platform with search, filter, sort, and modal CRUD.
- **Protected MongoDB API:** /api/v1/tasks – Endpoints protected by auth middleware enforcing multi-user data isolation.

3. Implementation Files & Technical Architecture

Component / File Path	Key Technical Functionality
server/controllers/task5Controller.js	RESTful API CRUD controller enforcing standard status codes (200, 201, 400, 404).
public/js/task5-api.js	Asynchronous fetch() client handling CRUD, loading spinners, and error alerts.
server/config/db.js	MongoDB Mongoose connection manager with zero-config memory-server fallback.
server/models/User.js	User model with bcrypt password hashing pre-save hook and toJSON credential sanitize.
server/models/Task.js	Task model with user reference (ObjectId) and compound indexing.
server/middleware/authMiddleware.js	Session & JWT auth guards and strict 403 Forbidden task ownership checks.
server/controllers/authController.js	User registration, login, logout, and token issuance.
server/controllers/taskController.js	Database CRUD controller enforcing multi-user isolation.
views/dashboard/index.ejs	Complete production dashboard template.
tests/test_task5.js & test_task6.js	Automated test scripts verifying full REST CRUD and multi-user isolation.

4. Submission Summary for Internship Portal

Cognifyz Level 3 (Advanced) Submission:

- Completed Tasks 5 & 6 in full compliance with Cognifyz internship guidelines.
- Created full CRUD RESTful API with uniform JSON envelopes and fetch() client.
- Integrated MongoDB with Mongoose, compound indexes, and zero-config embedded fallback.
- Implemented bcrypt password hashing and session/JWT authentication.
- Enforced multi-user isolation: Cross-user access rejected with HTTP 403 Forbidden.
- Verified with automated tests (test_task5.js and test_task6.js passed with code 0).